

Westphalian University of Applied Sciences, Gelsenkirchen  
Department of Computer Science and Communication

WISSENSCHAFTLICHE VERTIEFUNG INFORMATIK  
Scientific Capstone Project — Technical Report

# **Relational Alpha Factors with Graph Attention Networks**

A Dual-Track Study on US Equities and  
European Power Markets

**Wentao Ma**

`ma.wentao@studmail.w-hs.de`

M.Sc. Informatik

Supervisor: Prof. Laura Anderle  
Module: Wissenschaftliche Vertiefung Informatik, 12 ECTS  
Submitted: Gelsenkirchen, July 8, 2026

# Declaration of Originality

I hereby declare that I have written this report independently and without outside assistance, and that I have used no sources or aids other than those cited. All passages taken verbatim or in substance from published or unpublished work of others are identified as such. This report has not been submitted, in the same or a similar form, to any other examination authority.

Gelsenkirchen, July 8, 2026

Wentao Ma

# Abstract

Cross-sectional alpha factors score assets in isolation, yet the assets are coupled — statistically in equity markets, physically in European power markets. This report measures what that coupling structure is worth. To make the measurement interpretable, the information set is held fixed and two questions are separated: what a graph adds, measured against an unlearned uniform-propagation anchor, and what learned attention adds, measured as a graph attention network’s margin over that anchor on the same topology. One GAT kernel, one leakage-controlled evaluation harness, and one suite of research gates run on two structurally different graphs: an estimated correlation backbone over 49 US equities and the physical interconnector network over 20 European bidding zones.

On equities, learned attention beats unlearned propagation in 30 of 30 seeded runs; the best configuration reaches an out-of-sample rank-IC of  $0.0148 \pm 0.0043$  and long-short Sharpe of  $1.37 \pm 0.39$  over five seeds. The resulting composite factor is unique, consistent, and robust — three of four gates — but never beats the best single input factor. Attention analysis explains the modesty: the model applies a stationary, near-uniform reweighting of neighbour pooling, not sharp selection. On energy, the alpha framing failed in an instructive way: three nested evaluation artifacts — overlapping labels, wrong annualisation, and a clip on the evaluation return that concealed short-leg tail losses — each produced spectacular numbers before being caught by the project’s own controls, and under honest returns the strategy loses money. Reframed as forecasting, the energy track yields the cleanest relational evidence: the interconnector graph lifts node-level price forecast skill by +0.131 over an identically informed no-graph baseline, and on cross-border spreads — a target that does not exist without the relation — message passing beats a both-endpoint baseline in five of five seeds (+0.056 skill).

The report’s contributions are the isolation methodology, the two measured results, a worked three-artifact cautionary example for GNN-based market research, and the platform engineering that makes all of it reproducible from archived artifacts.

# Contents

|          |                                                                  |           |
|----------|------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                              | <b>5</b>  |
| <b>2</b> | <b>Background and Related Work</b>                               | <b>6</b>  |
| 2.1      | Alpha research in plain terms . . . . .                          | 6         |
| 2.2      | Cross-sectional alpha research . . . . .                         | 7         |
| 2.3      | Graph neural networks and attention . . . . .                    | 7         |
| 2.4      | Electricity price forecasting . . . . .                          | 7         |
| 2.5      | Where this report sits . . . . .                                 | 8         |
| <b>3</b> | <b>Platform Architecture</b>                                     | <b>8</b>  |
| 3.1      | The layered platform . . . . .                                   | 8         |
| 3.2      | Two seams instead of a fork . . . . .                            | 8         |
| 3.3      | From research run to warehouse mart . . . . .                    | 9         |
| <b>4</b> | <b>Methodology</b>                                               | <b>9</b>  |
| 4.1      | The canonical panel . . . . .                                    | 9         |
| 4.2      | Supervision label . . . . .                                      | 10        |
| 4.3      | Split protocol and leakage controls . . . . .                    | 10        |
| 4.4      | Model and training objective . . . . .                           | 11        |
| 4.5      | The four research gates . . . . .                                | 11        |
| 4.6      | The attention A/B: two no-learning anchors . . . . .             | 11        |
| 4.7      | Forecast-skill apparatus (energy reframe) . . . . .              | 12        |
| <b>5</b> | <b>Equity Track: GAT Relational Factors</b>                      | <b>13</b> |
| 5.1      | Controls before claims . . . . .                                 | 13        |
| 5.2      | First real-data run . . . . .                                    | 14        |
| 5.3      | Attributing the decay: graph versus model . . . . .              | 14        |
| 5.4      | Variance: seeds, devices, and the reporting protocol . . . . .   | 15        |
| 5.5      | Hyperparameter selection with clean hands . . . . .              | 15        |
| 5.6      | What the attention learned . . . . .                             | 16        |
| 5.7      | Summary of the equity evidence . . . . .                         | 16        |
| <b>6</b> | <b>Energy Track: From Alpha to Forecasting</b>                   | <b>16</b> |
| 6.1      | One kernel, two graphs — and a second negative control . . . . . | 17        |
| 6.2      | Real data, implausible result . . . . .                          | 17        |
| 6.3      | Redesign in the day-ahead frame . . . . .                        | 18        |
| 6.4      | The third artifact: a clip on the evaluation return . . . . .    | 19        |
| 6.5      | The reframe: does the graph improve forecasts? . . . . .         | 19        |
| 6.6      | Edge-level spread prediction: the relational payoff . . . . .    | 21        |
| 6.7      | Summary of the energy evidence . . . . .                         | 21        |
| <b>7</b> | <b>Discussion</b>                                                | <b>22</b> |
| <b>8</b> | <b>Conclusion and Future Work</b>                                | <b>23</b> |
|          | <b>Bibliography</b>                                              | <b>25</b> |
| <b>A</b> | <b>Reproducibility</b>                                           | <b>26</b> |

# 1 Introduction

Quantitative alpha research scores assets one at a time. A formulaic alpha reads a single stock’s prices and volumes and emits a number; the cross-section of those numbers, ranked, is the signal [1]. Yet the assets themselves are anything but isolated: equities co-move within sectors and supply chains, and European power prices are literally wired together by cross-border transmission lines. The premise of this project is that the coupling structure is information, and its subject is a careful measurement of how much.

The measurement question splits in two, and the split matters. That a graph-augmented model outperforms a non-relational baseline is easy to show and hard to interpret — the gain may come from the graph, from the extra parameters, or from information the baseline never saw. This project therefore fixes the information set (the same island alphas every model consumes) and separates two claims: what the *graph* adds, measured by an unlearned uniform-propagation anchor, and what *learned attention* adds, measured as the gap between a graph attention network [2] and that anchor on the same topology. Three research questions organise the work:

- RQ1 Does learned attention add value over unlearned relational propagation, holding inputs and topology fixed?
- RQ2 Does one GAT kernel and one evaluation harness transfer across structurally different graphs — an estimated correlation backbone over US equities and a physical interconnector network over European bidding zones — without manufacturing false positives?
- RQ3 On real power-market data, does the physical topology improve forecast skill, and where does its value concentrate?

A reader from outside quantitative finance should carry one picture through what follows (Section 2.1 defines the recurring vocabulary). Every model in this report is trained on the same task: score today’s cross-section of assets so that the scores rank the next period’s price moves, five trading days ahead for equities, one day ahead for power. Every model in a comparison also receives the same per-asset inputs: ten signals computed from each stock’s own prices and volumes on the equity track, and each zone’s price dynamics plus its published fundamentals (load, wind and solar forecasts among them) on the energy track. In particular, the graph neural networks here are not a vehicle for feeding extra data into price prediction — whatever exogenous drivers exist go to the graph-free baselines too. The relational models’ only privilege is topological: permission to exchange information along edges, estimated return correlations for equities and the physical cross-border grid for power. Every headline number in this report is therefore a margin, the value of that permission with everything else held fixed.

The answers, in brief. On the equity track, learned attention beats the uniform anchor in thirty of thirty seeded runs, and the best configuration reaches an out-of-sample (OOS) IC of  $0.0148 \pm 0.0043$  and Sharpe of  $1.37 \pm 0.39$  over five seeds; the resulting composite passes three of four research gates but never beats the single best island alpha, a failure this report states as plainly as the successes (Chapter 5). On the energy track the alpha framing collapsed honestly: three nested evaluation artifacts each produced spectacular numbers before the project’s own controls caught them, and under honest returns the strategy loses money (Chapter 6). Reframed as forecasting, the same track produced the project’s cleanest positives: the interconnector graph lifts node-level price forecast skill by +0.131 over a no-graph baseline with identical drivers, and on the irreducibly relational target — cross-border spreads — message passing beats a both-endpoint baseline in five of five seeds (Section 6.6).

Five contributions follow from this, each defended in a named part of the report. First, an isolation methodology for relational factors: matched no-learning anchors, four research gates, and structural leakage controls, defined once in Chapter 4 and reused unchanged by every experiment. Second, the equity result itself — attention’s margin over naive propagation as a distribution over seeds, devices, and configurations rather than a point estimate, together with an interpretability analysis showing the mechanism to be a gentle, stationary reweighting of mean pooling (Chapter 5). Third, a worked cautionary example: the energy alpha arc of Chapter 6, in which the project’s own controls catch three artifacts in sequence, ending in a verdict — rank correlation can be positive while the strategy loses money — that generalises beyond power markets. Fourth, the forecasting reframe and its control-validated graph value, concentrated on the relational target (Sections 6.5 and 6.6). Fifth, the platform engineering that makes the rest reproducible: two seams that let island and relational factors share one pipeline, warehouse marts built from research runs, and a test suite that pins the leakage-critical logic (Chapter 3).

A note on standards. This is a Master’s capstone report, and the working assumption throughout is that its most valuable output is not a headline Sharpe but a defensible measurement process. Wherever a result died under scrutiny, the scrutiny and the death are in the text; the negative results are load-bearing, not confessional.

The report proceeds as follows. Chapter 2 places the work in the alpha-research, GNN, and electricity-forecasting literatures. Chapter 3 describes the platform and the two seams. Chapter 4 fixes the measurement apparatus. Chapters 5 and 6 report the two tracks. Chapter 7 confronts threats to validity, and Chapter 8 closes with what a successor project should do differently.

## 2 Background and Related Work

### 2.1 Alpha research in plain terms

This report uses a small vocabulary that is standard in quantitative finance and opaque outside it; this section defines it once, so that the rest of the report does not presuppose a finance background.

An *alpha factor* is a rule that assigns every asset a score at every point in time, constructed so that higher-scored assets should subsequently outperform lower-scored ones. The prediction is about *relative* performance within a fixed universe, not about price levels: the object of interest is the *cross-section* — all assets observed at one timestamp — and whether today’s ranking of scores anticipates the ranking of returns realised over the following days. A factor that scores an asset from that asset’s own history alone (five-day price reversal, abnormal trading volume) is an *island factor* in this report’s terminology; a *relational factor* may also read the asset’s neighbours in a graph.

Two measurements recur on every page of Chapters 5 and 6. The *information coefficient* (IC) is the correlation between the scores issued at a timestamp and the returns realised afterwards; this report uses the rank form (*rank-IC*, a Spearman correlation), computed cross-sectionally at each timestamp and averaged. A rank-IC of 1 would be a perfect ordering; 0 is no predictive skill at all. Real factors are weak — the best out-of-sample composite in this report reaches a rank-IC of about 0.015 (Chapter 5) — and factors of this strength are useful only in combination, which is why the field builds them in large numbers. The *Sharpe ratio* evaluates a factor as a trading strategy instead: buy the top-ranked assets, sell short the bottom-ranked ones (a *long-short* portfolio, whose exposure to the overall market direction largely cancels), and divide the strategy’s mean return by its volatility, annualised. The two numbers can disagree, because ranking well says nothing about the magnitudes of wins and losses; Chapter 6 turns exactly this disagreement into one of the report’s main findings.

“Alpha” itself, in the industry idiom this report inherits from Kakushadze’s catalogue of practitioner formulas [1], names nothing deeper than a signal expected to carry predictive information beyond passive market exposure. No claim below depends on a stricter definition.

## 2.2 Cross-sectional alpha research

The equity track’s island factors follow the formulaic-alpha idiom that Kakushadze documented from industry practice: short arithmetic expressions over price and volume, evaluated per stock and ranked across the cross-section each day [1]. Individually weak and mutually correlated, such alphas are meant to be combined, and the interesting research questions live in the combination step. Machine learning has a measured track record here: Gu, Kelly and Xiu showed on six decades of US data that flexible models — trees and shallow networks in particular — roughly double the predictive  $R^2$  of linear benchmarks for cross-sectional returns, driven largely by interaction effects that linear factor models cannot express [3]. The same literature is emphatic about how easily such gains evaporate under sloppy evaluation. López de Prado catalogues the failure modes — label overlap, missing embargoes, selection on test data, backtest overfitting — and the purged, embargoed validation splits he advocates [4] are the direct ancestors of the split protocol in Section 4.3. This report treats that methodological literature as load-bearing: two of the energy track’s three artifacts (Section 6.2) are textbook instances of exactly the pathologies it names.

## 2.3 Graph neural networks and attention

Graph neural networks compute node representations by aggregating information from graph neighbourhoods. In the graph convolutional network of Kipf and Welling, the aggregation weights are fixed by the graph’s degree structure [5]; graph attention networks replace them with weights computed from the node features themselves, so the model learns which neighbours matter for each node [2]. Brody, Alon and Yahav later showed that the original formulation computes a restricted, effectively static form of attention and proposed GATv2, which reorders the operations to make the attention genuinely conditional on the query node [6]. Both operators are available off the shelf in PyTorch Geometric [7]; the alpha track uses the original operator, the forecasting work the GATv2 variant (Section 4.4).

Applying relational models to stock prediction is an established idea. Feng et al. combine a temporal encoder with relational graphs over company links and learn to rank stocks by return [8], and much subsequent work follows the same pattern: propose an architecture, report that it beats non-relational baselines. This project asks a deliberately narrower question. Rather than demonstrating that some graph-augmented architecture outperforms, it holds the information set fixed — the same island alphas every baseline consumes — and measures the marginal value of each ingredient separately: the graph via an unlearned uniform-propagation anchor, and learned attention via the gap between the GAT and that anchor (Section 4.6). The price of the narrower question is weaker headline numbers; the payoff is that when the equity track reports a positive attention margin in thirty of thirty runs (Chapter 5), the claim isolates a mechanism rather than a model.

## 2.4 Electricity price forecasting

Day-ahead electricity prices are set in a daily auction and driven by fundamentals — load, wind and solar in-feed, fuel prices — with properties equity researchers rarely face: strong diurnal seasonality, negative prices, and fat-tailed scarcity spikes. Weron’s review maps the field’s method families and stresses the role of these fundamentals and of rigorous out-of-sample evaluation [9]; Lago et al. sharpen the evaluation point for the deep-learning era, offering an open benchmark and best practices — among them, always comparing against naive and well-tuned simple baselines — precisely because the literature is rich in models that beat only weak straw men [10]. The forecast-skill ladder of Section 4.7, with persistence pinned at zero skill and a ridge on fundamentals as the serious no-graph baseline, is this project’s implementation of that advice. Its energy-specific contribution is not a new forecasting architecture but a controlled measurement of one factor the standard benchmarks do not isolate: whether the physical interconnector topology itself

adds forecast skill, at the node level and — where the answer turns out to be strongest — for cross-border spreads (Section 6.6).

## 2.5 Where this report sits

Against this background, three commitments distinguish this report from the related work above, which rarely combines them. Every relational claim is measured against matched no-learning anchors on identical inputs, so a positive result isolates an ingredient rather than crediting a whole architecture. One GAT kernel and one evaluation harness are carried across two structurally different graphs — an estimated correlation backbone over equities and a physical transmission network over bidding zones — with the negative controls travelling to both. And the honesty standard of the methodological literature is applied without exception: negative results (a failed value-added gate, a money-losing energy strategy, a null congestion hypothesis) are reported as findings, because the diagnostic arc that produced them is the part of this work most likely to help a reader.

# 3 Platform Architecture

The research of Chapters 5 and 6 runs on a data-engineering platform that was prior work to this project; the project’s architectural contribution is the pair of seams that let relational factors join that platform without forking it, and the loop that carries research outputs back into the warehouse. This chapter describes the platform in the depth the later chapters rely on — enough to make every named component findable in the repository — and no further.

## 3.1 The layered platform

The platform follows a conventional analytics layering. Ingestion fetches daily equity bars (Yahoo Finance), deterministic synthetic prices for offline work, and the ENTSO-E power-market series used in Chapter 6: day-ahead prices, load and renewable forecasts, realised load, generation mix, and physical cross-border flows with transfer capacities [11]. Raw and processed data land as Parquet files in a local data lake. The warehouse is DuckDB [12], chosen for zero-infrastructure local reproducibility — anyone with the repository can rebuild every table. Transformations are split by nature: factor mathematics lives in Python, while warehouse marts are dbt models [13] so that derived tables are declarative, tested, and documented. A Kestra flow orchestrates the daily equity and hourly energy pipelines, and a Streamlit dashboard is the read-only application layer.

One boundary is stated in the architecture documentation and repeated here: the platform is a research and portfolio-demonstration system. It places no orders, manages no broker state, and offers no investment advice.

## 3.2 Two seams instead of a fork

The capstone’s design problem was to add a second *family* of factors — relational ones — to a platform whose registries, apply step, backtests, and marts all assumed island factors. The naive solution is a parallel pipeline; the project’s solution is two narrow interfaces, fixed in architecture decision records before implementation.

The first seam is the propagator: one snapshot’s node features and topology in, one factor value per node out. Both the unlearned baseline (uniform neighbour averaging) and the trained GAT are adapters on this single interface. The seam is deliberately per-snapshot and transform-only — training happens elsewhere, weights are bound at construction — which keeps it a pure, testable function and keeps attention behind the interface rather than in it. The design consequence



carried through both result chapters: because baseline and GAT share one seam, the attention A/B of Section 4.6 is a one-line adapter swap, not a comparison across two code paths that might differ in untracked ways. A characterisation test pins the equivalence: the uniform propagator on a fully-connected topology must reproduce the platform’s pre-existing cross-market mean exactly.

The second seam unifies factor declaration. The two tracks had grown two factor idioms — equity as metadata plus a compute callable, energy as metadata interpreted by a monolithic imperative function. One `Factor` contract over the canonical *(time, entity)* panel, plus a `FactorProvider` interface for anything that yields factors, absorbed both: the energy legacy path is wrapped as one provider, and the graph factor is simply another, whose compute closes over a propagator and a topology source and runs the per-snapshot loop internally. At the registry and apply layer, island and relational factors are indistinguishable — which is precisely what makes the four-gate comparison of Section 4.5 a fair one, since every factor reaches the evaluation through the same code.

### 3.3 From research run to warehouse mart

Research results do not live in notebooks. A persistence step flattens a GAT run into four warehouse tables — the factor panel (composite, anchors, and island alphas per date and symbol), per-alpha diagnostics, the four gates as a one-row report, and the attention A/B as another — and dbt models build the presentation layer on top: a staging view over the panel, a tiered factor-versus-baseline mart, and a one-row scorecard joining gates to the A/B. The dashboard reads the marts, not the research code. The practical effect showed up during this report’s own preparation: rebuilding the marts end-to-end (21 dbt models and tests on the equity warehouse, 12 on the energy warehouse) is a single command per project, and the full test suite — 115 tests, of which 55 belong to the capstone modules — is the reproducibility contract for everything the previous chapters claimed. The leakage-critical pieces (labels, splits, rank-IC) are deliberately pure pandas so they are testable without the GNN dependency stack.

## 4 Methodology

Both research tracks run through one fixed measurement apparatus. This chapter defines it once: the data representation (Section 4.1), the supervision label (Section 4.2), the split protocol and its leakage controls (Section 4.3), the model and training objective (Section 4.4), the four research gates (Section 4.5), and the attention A/B that carries the core thesis test (Section 4.6). Section 4.7 adds the forecast-skill apparatus used after the energy track was reframed from alpha research to forecasting. Every experiment in Chapters 5 and 6 reports against this apparatus without modification; where a design choice was made in response to an observed failure, the chapter that reports the failure also documents the change.

### 4.1 The canonical panel

All data enters the pipeline as a panel indexed by *(time, entity)*: *(date, symbol)* on the equity track, *(timestamp, bidding zone)* on the energy track. The entities are the nodes a graph later connects. A *snapshot* is the cross-section of one timestamp — the unit that graph propagation operates on; no operation ever sees the panel as a whole time series.

Node features are deliberately not raw market data. Each track already maintains a registry of island factors — alphas that score a node from its own history alone, ten on the equity track and eight on the energy track. Their cross-sectionally ranked values form the feature matrix. This choice pins the research question: any value the graph model adds must come from relating nodes, because the per-node information is the same information the island baselines consume.

Concretely, the equity ten are functions of each stock’s own prices and volumes alone (reversal, medium-horizon momentum, intraday range position, volume shocks, overnight-gap behaviour); nothing exogenous enters that track. The energy eight mix each zone’s price dynamics with its published fundamentals, such as residual load, wind and solar forecast terms, demand surprise, and the gas spark spread. Which signals enter matters less than the rule that they enter everywhere: island baselines, anchors, and GAT consume this identical matrix, so a relational model can hold no information advantage over the baselines it is measured against. Missing values are filled with the per-snapshot median, which in rank space is the no-information position; a node with no data is indistinguishable from a node ranked exactly in the middle.

## 4.2 Supervision label

The label is the forward return over a fixed horizon  $k$ , standardised per snapshot so that the model learns to rank the cross-section rather than to predict absolute magnitudes. On the equity track it is  $r_{t,k} = p_{t+k}/p_t - 1$  with  $k = 5$  trading days, z-scored within each snapshot. On the energy track it is  $r_{t,k} = (p_{t+k} - p_t) / \max(|p_t|, 20)$  with  $k = 24$  hours, clipped to  $\pm 0.8$  and z-scored; the denominator floor is unavoidable because European power prices cross zero and turn negative, so a plain relative return divides by numbers arbitrarily close to zero.

The label is supervision only. It is computed from prices at  $t + k$  and never appears as a feature; the last  $k$  snapshots of each entity have no label and are excluded from training. The energy clip, introduced for label stability, later turned out to have a second life as an *evaluation* artifact — Chapter 6 reports how clipping the evaluation return concealed tail losses, and why the shipped evaluation is unclipped.

## 4.3 Split protocol and leakage controls

Leakage was treated as the primary methodological risk from the start, and the controls are structural rather than procedural: they are asserted in code, not remembered by the experimenter.

A single in-sample/out-of-sample boundary is placed at 70% of the timestamps. That one date is computed once and passed explicitly to every consumer: graph construction builds edges only from data strictly before it, model selection sees only data inside it, and the gate evaluation scores only data after it. The alternative — letting each component derive its own boundary — failed quietly during development: three components each assumed 70% and happened to agree, so changing any one ratio would have leaked without any test failing. Centralising the date turned that silent coupling into an explicit contract.

Within the in-sample region the layout is

$$\text{train} \mid \text{embargo}(k) \mid \text{valid} \mid \text{embargo}(k) \mid \text{OOS},$$

with the validation block embargoed on both sides. The leading embargo is standard practice: the last training labels reach  $k$  steps forward and must not overlap the first validation features [4]. The trailing embargo is easy to omit and was nearly omitted here: validation labels also reach  $k$  steps forward, so without it, best-epoch selection would be scored partly on returns that belong to the out-of-sample window. A dedicated assertion (`is_constrained_split`) rejects any split violating this layout.

Two automated controls argue that the pipeline as a whole produces no false positives. A negative control shuffles the labels and asserts that validation IC collapses to zero; a positive control plants a synthetic signal and asserts that training recovers it. Both run in the test suite on every change to the leakage-critical code, which is deliberately pure pandas — testable without a GPU or the torch dependency.

## 4.4 Model and training objective

The relational model is a stacked graph attention network [2] with two layers, implemented on PyTorch Geometric [7]. (The later forecasting work of Chapter 6 uses the GATv2 variant [6]; the alpha track predates it in the codebase and keeps the original operator.) The model maps one snapshot’s node features and topology to one score per node; scores across snapshots form the relational composite factor.

Training minimises an IC loss — the negative Pearson correlation between predictions and labels within each snapshot — a differentiable stand-in for the evaluation metric, the per-snapshot rank-IC (the Spearman correlation of predictions and labels, reported throughout as the OOS IC). An MSE variant is retained for comparison; Chapter 5 reports the probe that motivated the switch, in which the IC loss recovered a planted signal far better than MSE. Model selection is by best validation IC with epoch checkpointing: the weights deployed are those of the best validation epoch, not the last one. This detail matters in practice — on the first real-data run the validation IC peaked mid-training and then decayed to roughly half its peak, so returning the final epoch would have discarded most of the signal.

Two deployment modes are evaluated throughout. *Single* fits once on the in-sample region and scores the entire out-of-sample window with frozen weights. *Walk-forward* refits at every 63-snapshot boundary through the out-of-sample window, each refit using only data whose labels predate its boundary, with the same embargoed layout per fold. Every walk-forward score therefore comes from a model that would have been trainable at that date in deployment. All headline runs are CPU; a GPU benchmark found no speedup at this graph size (about 50 nodes).

## 4.5 The four research gates

A relational composite is only interesting if it survives comparison against the island factors it was built from. Four gates, all scored on the out-of-sample slice, formalise that comparison (Table 4.1).

Table 4.1: The four research gates. The composite is compared against the single island alphas (ten equity, eight energy), out-of-sample.

| Gate        | Definition                               | Pass threshold   |
|-------------|------------------------------------------|------------------|
| Value-added | composite OOS Sharpe vs. best single     | composite > max  |
| Consistency | IS/OOS IC same sign, magnitude retention | score $\geq 0.5$ |
| Uniqueness  | max  Spearman  vs. each single           | < 0.7            |
| Robustness  | blend: OOS coverage, IC IR, drawdown     | score $\geq 0.5$ |

The value-added gate is deliberately the strictest formulation — beating the maximum of  $N$  competitors is a higher bar than beating their mean, and a composite can be genuinely useful while failing it. Results therefore also report the beat-rate against the full set of singles, and Chapter 5 discusses the one gate this project never closed.

## 4.6 The attention A/B: two no-learning anchors

A GAT that beats the island factors has not yet demonstrated that *learned attention* was the ingredient that mattered; the improvement could come from neighbour averaging alone. Every run therefore carries two anchors, computed from the same inputs and, where applicable, the same topology: the *island anchor*, an equal-weight composite of the input alphas with no propagation at all; and the *uniform anchor*, the same composite averaged uniformly over the GAT’s own topology — relational, but with nothing learned. Figure 4.1 draws the arrangement: three factor constructions, one input matrix, one harness.

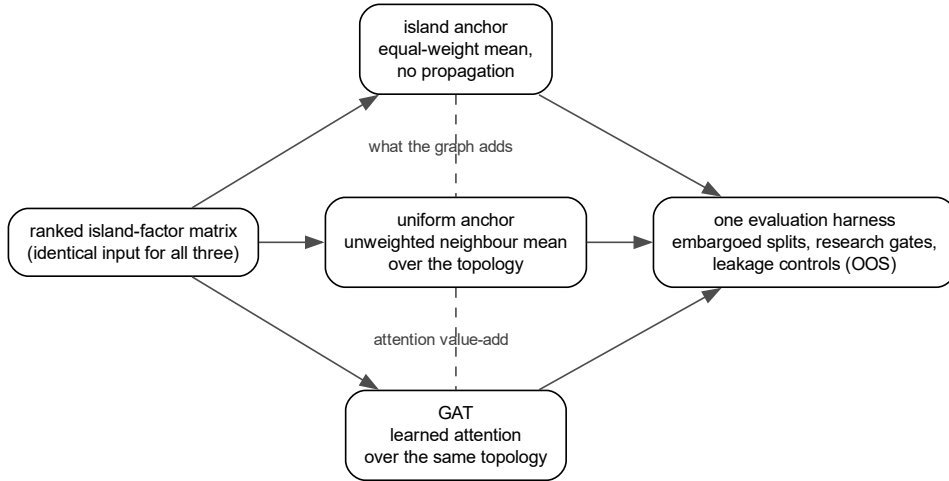


Figure 4.1: The attention A/B. All three factor constructions read the same ranked island-factor matrix and are scored by the same evaluation harness; the two dashed comparisons are the quantities this report is about — what the graph adds (island anchor versus uniform anchor) and what learned attention adds (uniform anchor versus GAT, the attention value-add).

The difference between the GAT’s out-of-sample Sharpe and the uniform anchor’s is the *attention value-add*, the quantity the core thesis stands on. It isolates the contribution of learned attention while holding the information set and the graph fixed. Where the four gates ask “is the composite a good factor?”, the A/B asks the sharper question “did learning the edge weights matter?” — and the two questions turn out to have different answers in Chapter 5.

## 4.7 Forecast-skill apparatus (energy reframe)

After the energy alpha attempt concluded negatively (Chapter 6), the energy question was reframed from “is there a tradeable cross-sectional signal?” to “does the physical interconnector graph improve forecast accuracy?”. The measurement apparatus changes accordingly. The metric is a skill score against persistence,

$$\text{skill} = 1 - \text{MSE} / \text{MSE}_{\text{persistence}},$$

scored out-of-sample at  $k = 24$  hours, alongside MAE and the cross-sectional rank-IC. Persistence (tomorrow equals today) scores zero by construction; a model must beat carrying yesterday forward to show any skill at all.

Candidates form a ladder in which each rung isolates one ingredient: a seasonal-naive rung isolates the diurnal cycle; a per-node ridge regression on physical drivers (load, wind and solar forecasts, fuel price) isolates what a zone’s own fundamentals add; a ridge with uniform neighbour averaging isolates what the interconnector graph adds without learning; and the GAT rungs isolate what learned attention adds on top. The anchors of Section 4.6 reappear here in forecasting form — the value claim is the margin over the no-graph rung *and* over the uniform-graph rung, not the absolute skill.

Feature timing follows the day-ahead market’s information structure: load, wind, and solar forecasts for hour  $t + k$  are published before  $t$  and may legitimately be used as features for  $t + k$ ; realised quantities (actual load, generation mix) are known only up to  $t$  and enter as point-in-time anchors. The synthetic negative control carries over: synthetic zones are generated independently,

so the test suite asserts that the graph rung does not beat the no-graph rung on synthetic data. Any measured graph lift can then only come from real cross-zonal coupling.

## 5 Equity Track: GAT Relational Factors

The equity track asks whether propagating island-alpha information over a correlation graph produces a factor worth having, and — the sharper question, RQ1 of Chapter 1 — whether *learning* the propagation weights matters. This chapter reports the experiments in the order they constrained each other: the controls that license any positive claim (Section 5.1), the first real-data run (Section 5.2), the ablation that attributed its weaknesses (Section 5.3), the seed and device studies that set the reporting discipline (Section 5.4), hyperparameter selection under that discipline (Section 5.5), and finally what the trained attention actually does (Section 5.6). Table 5.1 states the headline up front.

Table 5.1: Equity track headline results (real yfinance data, 49 names, IC loss).

| metric                             | value                                        | note                                                          |
|------------------------------------|----------------------------------------------|---------------------------------------------------------------|
| Gates passed                       | 3 / 4                                        | Uniqueness + Consistency + Robustness pass; Value-added fails |
| Attention value-add                | 30/30 seeds positive                         | vs the uniform anchor (no learned attention)                  |
| Composite best-epoch validation IC | 0.0756                                       | best-by-valid-IC checkpoint selection                         |
| Composite OOS Sharpe               | 1.42                                         | vs best single 2.88 (beats 9 of 10 single factors)            |
| Graph                              | correlation top-k ( 50 names + GICS sectors) | estimated backbone (not physical)                             |
| Data                               | real yfinance<br>2021-01→2026-06             | 49 names; IC loss; 50 epochs                                  |

### 5.1 Controls before claims

No positive number in this chapter means anything unless the pipeline can be shown to find nothing where nothing exists. Three controls establish this.

First, the synthetic negative control: fifty random-walk price series run through the full pipeline — graph construction, training, gates. The gates correctly report no edge. Value-added and Consistency fail (OOS IC around  $-0.06$ , negative Sharpe value-add), while Uniqueness and Robustness pass, as they should: they measure structural properties of a factor, not the presence of signal.

Second, the automated pair of controls that runs in the test suite on every change. Shuffling each snapshot’s labels across nodes and retraining must collapse the validation IC to zero, and it does, for both loss functions. Planting a recoverable signal — the label set equal to a standardised input feature — must let the same training loop recover it, and it does. The second half is what makes the first half meaningful: the IC collapses on shuffled labels because there is nothing to learn, not because the trainer is broken.

Third, a convergence probe for the training objective itself. On the planted-signal panel the IC loss recovers the signal essentially completely (validation IC 0.99 and above at learning rate  $10^{-3}$ ), where MSE plateaus at 0.33. The IC loss is unstable early at higher learning rates — at  $5 \times 10^{-3}$

it bounces near zero for some forty epochs before converging — which fixed the default learning rate at  $10^{-3}$  and the IC loss as the default objective.

## 5.2 First real-data run

The first run on market data used daily bars for the fifty-name universe (one name, ORCL, failed to download, leaving 49 names, 1,364 trading days and 66,836 panel rows spanning January 2021 to June 2026), the ten island alphas as node features, a static correlation graph built from data before the split date, and IC-loss training for fifty epochs.

Training behaved like the textbook picture. The training loss fell monotonically while the validation IC rose to a peak of 0.0756 at epoch 32 and then decayed to 0.0418 by the final epoch — plain overfitting. Best-epoch checkpointing therefore deployed the epoch-32 weights; had the pipeline returned the final epoch, as an earlier version of the code silently did, the deployed model would have carried barely half the validation IC. The checkpoint fix paid for itself on its first contact with real data.

The four gates split three to one. Consistency (score 0.63), Uniqueness (maximum absolute correlation against any single alpha, 0.257) and Robustness (0.68) pass; Value-added fails, with a composite OOS Sharpe of 1.42 against 2.88 for the best single island alpha. The honest reading cuts both ways. The composite is a real, unique, consistent signal — the second-best of the eleven factors on the board, beating nine of the ten singles it was built from, and correlated with none of them above 0.26. It is genuinely new information rather than a repackaging of its inputs. But the strictest bar — beat the maximum of ten competitors — stayed open here and, as the rest of this chapter shows, never closed.

The run also posed the question that drove the next experiment: the validation IC of 0.076 decayed to an OOS IC of 0.007 over the roughly 1.6-year OOS window. Was the frozen correlation graph going stale, or the frozen model weights?

## 5.3 Attributing the decay: graph versus model

A  $2 \times 2$  ablation crossed the graph axis (static, frozen at the split date, versus dynamic, rebuilt per snapshot from a trailing window) with the retraining axis (single fit versus walk-forward refits every 63 snapshots). All four cells used one shared data fetch (all fifty names this time, 68,200 rows), the same split, the same seed, and carried the two no-learning anchors of Section 4.6. Table 5.2 shows the result.

Table 5.2: The graph  $\times$  retraining ablation, one seed, IC loss. “Att. VA” is the attention value-add: GAT OOS Sharpe minus the uniform anchor’s. Anchors, for reference: island mean Sharpe  $-2.13$ ; uniform anchor  $-1.05$  (static graph) and  $-1.19$  (dynamic).

| Configuration          | OOS IC    | OOS Sharpe | Gates   | Att. VA |
|------------------------|-----------|------------|---------|---------|
| static + single        | $-0.0051$ | 1.04       | F/F/T/T | 2.09    |
| dynamic + single       | $-0.0348$ | $-1.01$    | F/F/T/T | 0.18    |
| static + walk-forward  | 0.0200    | 1.95       | F/T/T/T | 3.00    |
| dynamic + walk-forward | 0.0007    | 0.05       | F/T/T/T | 1.24    |

Two findings, one of them negative. Walk-forward retraining is the lever: it lifts the static-graph cell from an OOS IC of  $-0.005$  to  $+0.020$  and flips the Consistency gate to pass. The decay diagnosed in Section 5.2 was model staleness. The dynamic graph, by contrast, hurts in both arms — per-snapshot correlation estimates from a trailing window are noisy, and the frozen train-period graph evidently acts as a regulariser. This negative result is worth having: the intuitively appealing refinement (“keep the graph fresh”) makes things worse as implemented, and

the project kept the static graph with its mild, disclosed in-sample lookahead rather than trade OOS performance for cosmetic cleanliness.

The attention A/B, meanwhile, is positive in every cell. Both no-learning anchors are clearly negative out-of-sample; the GAT beats the uniform anchor by +0.2 to +3.0 Sharpe on the same topology with the same inputs, and its scores correlate with the uniform anchor’s at Spearman  $-0.29$  to  $-0.14$  — it is not learning to smooth, it is learning something else. Against the strict value-added bar the best cell still falls short (best single at 3.07; the gap narrows from  $-2.03$  to  $-1.12$ ), but the core-thesis comparison holds regardless of which cell one reads.

One warning surfaced here as well: the static+single cell read  $-0.0051/1.04$  where the near-identical first run had read  $+0.0066/1.42$ . Run-to-run variance was clearly material, which is what the next section is about.

## 5.4 Variance: seeds, devices, and the reporting protocol

Five seeds per arm on the same panel put numbers on the spread. The single-fit arm scored an OOS Sharpe of  $0.73 \pm 0.25$  (per-seed range 0.36–1.04) and the walk-forward arm  $1.15 \pm 0.75$  (range 0.13–1.95); walk-forward’s mean advantage came with three times the variance, and a Welch  $t$  of about 1.2 at  $n = 5$  per arm. The single-seed reading of the ablation — “walk-forward is the lever” — had ridden the luckiest walk-forward draw. Meanwhile the attention value-add was positive in all ten runs, spanning +1.18 to +3.00 Sharpe against a seed-independent anchor: the core claim survived the variance study; the walk-forward claim was downgraded to “improves the mean, adds variance, not significant at  $n = 5$ ”.

A hardware benchmark added an unplanned second lesson. Moving training to a laptop GPU was 24–32% *slower* at this graph size — fifty nodes and ten features per snapshot is latency-bound, and kernel-launch overhead dominates — so production runs stayed on CPU. More interesting was that the same seed on a different device swung the result by the full width of the seed distribution. The seed fixes the initial weights, but dropout masks come from each device’s own random stream and parallel reductions round in a different order, so training diverges from the first step; a cross-device rerun is another draw from the same distribution, not the same experiment on new hardware.

Together these fixed the reporting protocol for everything that follows: every headline number is a mean  $\pm$  standard deviation over at least five seeds on a fixed device, and what the project claims to be reproducible is the distribution, not bit-exact point estimates.

## 5.5 Hyperparameter selection with clean hands

The hyperparameter search was designed so that the OOS window could not influence selection even in principle. Twenty-four configurations (learning rate  $\{5 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$ , hidden size  $\{32, 64\}$ , heads  $\{2, 4\}$ , layers  $\{1, 2\}$ ) ran three seeds each — 72 training runs scored exclusively on the double-embargoed validation IC, with no OOS metric computed anywhere in the grid. Only the winner then touched the OOS window, once, with fresh seeds.

The grid’s surface is flat on top: six configurations sit within validation IC 0.10–0.115, and the shipped default ranked third. The one structural regularity is depth — all six top configurations have two GAT layers. Results are therefore robust to reasonable hyperparameter choices, which is simultaneously a robustness point and a caution against reading much into tuning.

The winner’s OOS validation initially produced a finding that did not survive its own scrutiny. The validation run had silently executed on the GPU — the CUDA wheel had just been installed and the fitting routine auto-selected it — while the reference numbers it was compared against were CPU runs. Given the cross-device divergence of Section 5.4, that comparison confounded device with configuration. The code was fixed the same day (CPU is now an explicit default, CUDA opt-in) and the validation rerun on CPU. The deconfounded result: in the single arm the winner is a wash against the default (0.80 vs. 0.73 Sharpe, IC near zero in both), but in the

walk-forward arm it is better on the mean and far tighter — OOS IC  $0.0148 \pm 0.0043$  against the default’s  $0.0055 \pm 0.0147$ , a 2.7-fold mean with a third of the variance, all five seeds positive on both IC and Sharpe, and the IC mean 3.4 standard deviations above zero. Within this configuration, walk-forward versus single reaches significance (Welch  $t \approx 3.9$  on IC,  $\approx 2.4$  on Sharpe at  $n = 5$  per arm), upgrading the walk-forward claim to “significant in the tuned configuration, directionally consistent everywhere”.

The best known setup is therefore the static graph, IC loss, walk-forward refits, and the winner hyperparameters, at an OOS IC of  $0.0148 \pm 0.0043$  and OOS Sharpe of  $1.37 \pm 0.39$ . Across the three run families — seed study, GPU validation, CPU rerun — the attention value-add is positive in 30 of 30 seeded runs, spanning  $+0.69$  to  $+3.03$  Sharpe. That, not any single point estimate, is the equity track’s central result.

## 5.6 What the attention learned

The A/B says learned attention beats uniform averaging; attention analysis on a representative trained model (919,336 edge-weight rows over 1,364 dates) says how.

The attention is genuinely relational. The mean self-loop weight is 0.084 — the model places 91.6% of its attention mass on neighbours — which rules out the trivial failure mode in which attention collapses onto each node’s own features and the graph is decoration. But across a node’s roughly 12.5 neighbours the distribution is nearly uniform: normalised entropy 0.956, a top-neighbour share of 0.152 where exact uniformity would give 0.08, and a sector-homophily lift of just  $+0.019$  (attention-weighted same-sector share 0.330 against a structural 0.311). The trained model is a gentle reweighting of mean pooling, not a sharp selector. This is the mechanism behind the whole chapter: small, learned tilts on top of broad pooling explain why the GAT’s edge over the uniform anchor is consistent yet modest, and why its scores are nearly uncorrelated with the anchor’s.

The coarse structure is also temporally stationary: the self/neighbour split holds at 0.91–0.92 and the homophily lift stays within  $+0.01$  to  $+0.03$  across the full 2021–2026 span, with no visible break at the IS/OOS boundary (Figure 5.1). The project’s original design hypothesis — that attention would adapt to macro regimes — is not supported at this aggregate level, and the report states so; whatever adaptation exists lives in per-name weights inside a stable coarse structure. Finally, the names drawing the most incoming attention are mega-caps and sector bellwethers — AMZN, NFLX, AVGO, GS, CAT, NVDA, MSFT, LIN — exactly the names the rest of the universe co-moves with, a qualitative validity check the attention passes.

## 5.7 Summary of the equity evidence

Learned attention over the correlation graph reliably beats unlearned propagation over the same graph with the same inputs: positive attention value-add in 30 of 30 seeded runs across two hyperparameter configurations and two devices. The resulting composite is a real, unique, consistent factor that beats nine of its ten inputs but not the best of them; the value-added gate remains the one honest failure. Dynamic graphs hurt, walk-forward retraining helps (significantly so in the tuned configuration), single-seed and single-device numbers are not to be trusted, and the attention mechanism itself is interpretable — relational, nearly uniform, stationary, and centred on economically sensible hubs.

# 6 Energy Track: From Alpha to Forecasting

The energy track set out to repeat the equity result on a physically grounded graph: twenty European bidding zones connected by the actual cross-border transmission network, with day-



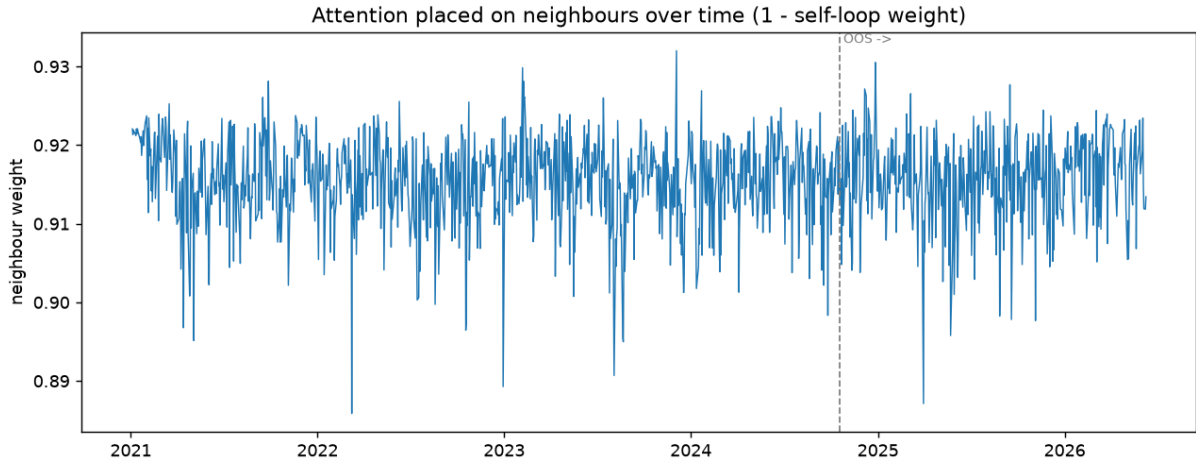


Figure 5.1: Attention mass placed on neighbours (one minus the self-loop weight) per snapshot over 2021–2026, from the archived attention analysis of the representative trained model. The share stays in a narrow band around 0.91–0.92, with no break at the dashed IS/OOS boundary — the stationarity reported in the text.

ahead spot prices from the ENTSO-E Transparency Platform [11]. Figure 6.1 shows that graph as configured in the repository — twenty zones, 38 physical borders and HVDC links, reference data rather than anything estimated from prices. The track ended somewhere more interesting than its plan. The cross-sectional alpha framing failed — not quietly, but through three nested evaluation artifacts, each of which initially produced spectacular numbers and each of which the project’s own controls caught (Sections 6.1 to 6.4). The track was then reframed as a forecasting problem, where the graph finally showed genuine, control-validated value, concentrated exactly where relational structure should matter most: on cross-border spreads (Sections 6.5 and 6.6). This chapter reports the arc in order, because the order is the contribution. Two research questions run through it: whether one kernel transfers to a physical graph without manufacturing false positives (RQ2), and whether that graph improves forecasts (RQ3).

## 6.1 One kernel, two graphs — and a second negative control

The dual-track design goal was engineering economy: the equity GAT, its training loop, its four gates, and its A/B anchors run unchanged on the energy panel; only the graph (physical interconnectors instead of estimated correlations), the label horizon (24 hours instead of 5 days), and the node set (bidding zones instead of symbols) differ. That goal was met literally — the same kernel functions are called by both tracks — and the first energy run doubled as a control.

On synthetic power data (2,857 hourly snapshots, 57,140 rows, generated without any cross-zonal coupling) the pipeline correctly finds nothing: OOS IC around  $-0.01$  in both retrain arms, and, crucially, the attention value-add is *negative* ( $-0.74$  single,  $-0.59$  walk-forward). When there is no relational signal, the GAT is worse than uniform neighbour averaging, not better. This is the energy analogue of the equity negative control, on a structurally different graph: even a learned attention layer over a physical topology does not manufacture signal from noise.

## 6.2 Real data, implausible result

The live ENTSO-E run covered all twenty zones for the full year 2024 at hourly resolution — 175,200 rows, with every configured zone returning data. Real power prices immediately justified the label’s engineering: they spanned  $-427$  to  $1,896$  EUR/MWh, negative in 3.95% of hours. The pipeline ran end to end without kernel changes. The numbers it produced were an OOS IC of

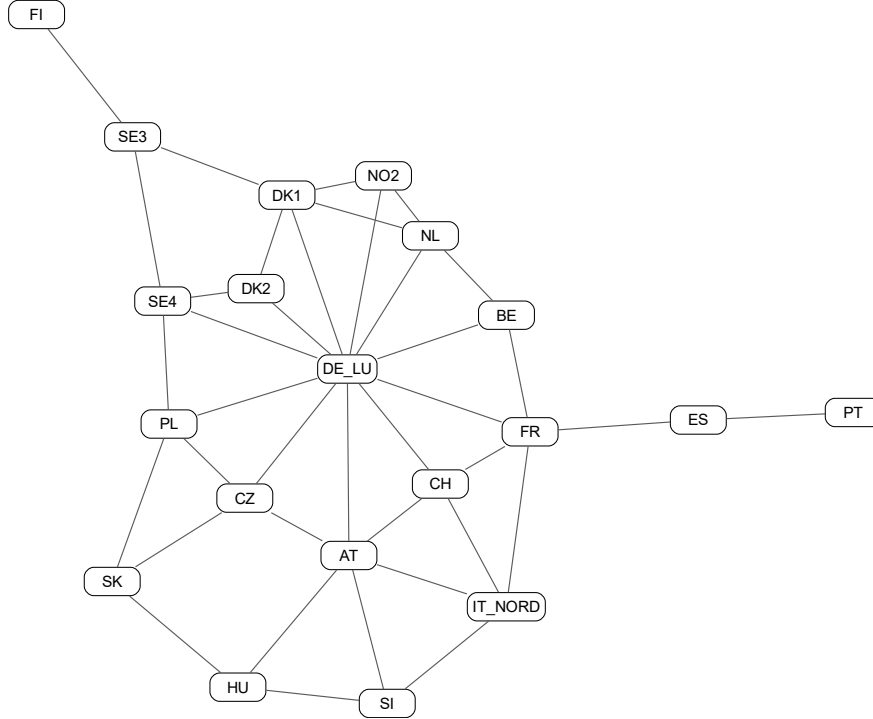


Figure 6.1: The energy graph: 20 European bidding zones and their 38 physical cross-border interconnections (land borders and HVDC cables), drawn from the repository’s reference data. Unlike the equity correlation graph, nothing here is estimated — the topology is the grid. DE\_LU, with eleven interconnectors, is the network’s natural hub.

0.20, an OOS Sharpe of 8.25, a composite roughly one hundred times its best single input, and four gates out of four.

The same kernel earns a Sharpe of about 1.4 on equities. A result this much better is not a discovery; it is a symptom. Diagnosis found three compounding artifacts. First, overlapping labels: a 24-hour forward return sampled hourly gives consecutive observations that share 23 of their 24 hours — measured lag-1 autocorrelation 0.91 — while the Sharpe computation treats them as independent, understating the denominator by an order of magnitude. Second, the “signal” is trivial: a one-line predictor, the negative of the current spot price, scores a cross-sectional IC of 0.235, *higher* than the GAT’s 0.20. Whatever the model learned, it is dominated by deterministic diurnal mean-reversion. Third, the annualisation constant was the equity one (252 periods per year) applied to hourly data (8,760). On top of all three sits a market-structure lookahead: the day-ahead auction publishes all 24 hours of day  $D$  at once on  $D - 1$ , so an “hourly” 24-hour-ahead label is partly contemporaneous information.

The point the report wants to preserve is that the magnitude itself was the alarm. The same scepticism that made the synthetic controls a design feature flagged the real-data number as not credible, rather than publishing a Sharpe of 8.

### 6.3 Redesign in the day-ahead frame

The redesign moved the evaluation to the market’s actual decision frequency. The day-ahead auction clears once per day, so the snapshot became the delivery day: base-load-aggregated features from data up to day  $D$ , a label equal to the genuinely-unknown next-day base-load price change  $D \rightarrow D+1$ , non-overlapping by construction, annualised at 365.

The mechanical checks all came back clean. Label lag-1 autocorrelation fell from 0.91 to  $-0.10$ ;

the shuffle-label control gave a validation IC of 0.07 against 0.19 for real labels, so the remaining signal is not leakage. And yet the result stayed implausible: GAT composite OOS Sharpe 11.08 — with the trivial negative-price predictor at 6.15. The interim reading was that the pipeline was now honestly measuring a real but non-tradeable physical regularity: the law of one price across interconnected zones, already priced by cross-border capacity auctions. That reading was half right, and its other half is the most instructive result in this report.

## 6.4 The third artifact: a clip on the evaluation return

A direct challenge — *have you actually confirmed the strategy would lose nothing?* — prompted one more check: recompute the same strategies under different definitions of the evaluation return. Table 6.1 shows what that found.

Table 6.1: The return-definition diagnostic (real ENTSO-E daily frame, OOS). The  $\pm 0.8$  clip, inherited from the training label, was also applied to the evaluation return in E13 — and manufactured the entire positive result.

| Evaluation return                     | trivial –price Sharpe | GAT Sharpe |
|---------------------------------------|-----------------------|------------|
| floored + $\pm 0.8$ clip (as trained) | 6.280                 | 11.000     |
| winsorised $\pm 0.5$                  | 4.610                 | —          |
| log return                            | 3.750                 | —          |
| plain relative return (honest)        | –1.520                | –1.540     |

The mechanism is specific and worth stating plainly. The long-short strategy shorts expensive zones. European power prices have fat right tails — scarcity spikes reaching 1,896 EUR/MWh in this sample — and the  $\pm 0.8$  clip on the evaluation return caps exactly those short-leg losses. Removing the clip turns a Sharpe-11 winner into a strategy that loses money, for the GAT and the one-line baseline alike. The rank-IC, meanwhile, stays at about 0.21 under both return definitions: cheap zones really do tend to rise relative to expensive ones, but ranking correctly is blind to tail magnitudes, and the tails are where this strategy’s money is lost. A positive IC coexisting with negative PnL is the transferable lesson — never transform the evaluation return, and never read rank quality as profitability. The equity track is unaffected: its evaluation return was never clipped.

The final verdict on the energy alpha question is therefore airtight and negative. Under honest returns the day-ahead cross-sectional strategy loses money (Sharpe about  $-1.5$ ); every positive energy number along the way was an artifact — overlap, annualisation, then the return clip — and the GAT adds no tradeable value over a one-line baseline. The fix (evaluation on unclipped returns) shipped as the default, so the repository can no longer reproduce the inflated numbers.

## 6.5 The reframe: does the graph improve forecasts?

What survives E13b is a genuine question the data can answer: not “is there alpha?” but “does the interconnector graph improve forecast skill?” — measured with the ladder of Section 4.7 on six months of real ENTSO-E data (20 zones, 87,360 rows, with day-ahead load, wind and solar forecasts, realised load, and generation mix as drivers). Table 6.2 shows the node-level ladder, and Figure 6.2 draws it with one added ingredient per rung.

Three findings, in decreasing order of robustness. First, the graph carries real forecast value: uniform neighbour averaging over the interconnector topology lifts skill by  $+0.131$  over the no-graph ridge with the same drivers. The synthetic negative control (independent zones) shows no such lift, so the gain is genuine cross-zonal coupling. This rung is pure numpy — no learned model is involved — which makes it the hardest number in the chapter to argue with.

Table 6.2: Node-level price forecast skill ladder ( $\text{skill} = 1 - \text{MSE}/\text{MSE}_{\text{persistence}}$ , OOS,  $k = 24\text{h}$ ).

| predictor           | skill | rank ic | note                                                    |
|---------------------|-------|---------|---------------------------------------------------------|
| persistence         | 0.000 | 0.637   | reference (tomorrow = today)                            |
| seasonal_naive      | 0.000 | 0.637   | diurnal carry                                           |
| no_graph_ridge      | 0.224 | 0.636   | own physical drivers (ridge)                            |
| uniform_graph_ridge | 0.355 | 0.584   | + interconnector-neighbour mean<br>(unlearned graph)    |
| gat_node            | 0.347 | 0.612   | learned attention (PyG GATv2;<br>5-seed mean)           |
| gat_congestion      | 0.346 | 0.615   | + price-spread congestion edge<br>feature (5-seed mean) |

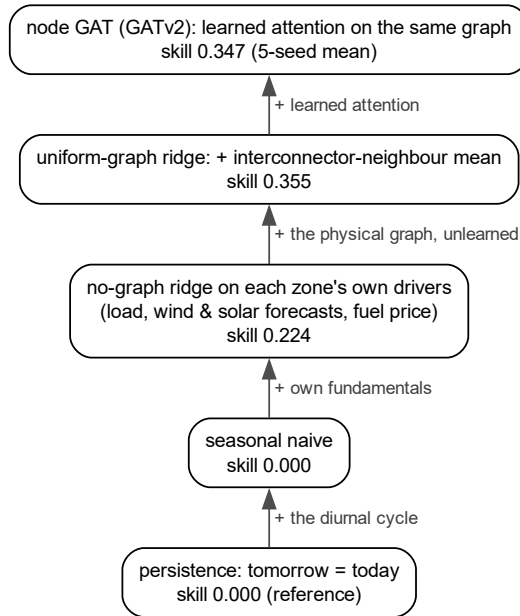


Figure 6.2: The node-level forecast ladder of Table 6.2, drawn as a ladder. Each rung adds exactly one ingredient, so each skill increment isolates one cause. The published drivers enter at the ridge rung and are shared by every rung above it; the graph’s contribution (+0.131) enters below any learning.

Second, learned attention beats the uniform anchor on cross-sectional *ranking* in five seeds of five (rank-IC 0.61 against 0.58) but is a wash on MSE skill (better in two seeds of five). Uniform neighbour-averaging turns out to be near MSE-optimal on this graph — a hard bar — so the GAT’s robust edge here is order, not magnitude, echoing the equity finding that attention’s value is a modest tilt on broad pooling.

Third, the congestion hypothesis — that attention should down-weight saturated borders if told how congested they are — is not supported. Two operationalisations of the edge feature were tried: the current cross-border price spread (a leak-safe proxy), and ground-truth physical flows over day-ahead transfer capacity fetched for all 38 borders. The proxy’s apparent gain existed only in a hand-rolled dense GAT implementation (+0.031 skill, five of five seeds) and vanished under the standard PyG GATv2 (−0.002, two of five); the ground-truth version beat node-only attention in one seed of five. The hypothesis is not refuted — capacity data covered only 36% of borders, and the most coupled region uses flow-based coupling that a flow-over-capacity ratio cannot express — but as a smooth additive edge feature, congestion does not add forecast skill. The two-backend check deserves its methodology note: a single-seed PyG run had initially shown +0.072, a lucky draw that a five-seed rerun corrected, and the cross-implementation comparison is what stopped a fragile effect from becoming a claim.

## 6.6 Edge-level spread prediction: the relational payoff

The last experiment moved the target instead of the model. A cross-border price spread  $\text{spot}_a - \text{spot}_b$  is irreducibly relational — it does not exist for a single node, and it is the object that transmission rights actually price. A GAT computes node embeddings by message passing over the interconnector graph; a small MLP head predicts each border’s next-day spread from the two endpoint embeddings and the current spread. The baseline that matters is a ridge that already sees both endpoints’ drivers and the current spread — so any gain over it is whole-network context, not information the baseline lacks. Table 6.3 shows the ladder over the 38 borders.

Table 6.3: Edge-level cross-border spread prediction (38 borders, OOS,  $k = 24\text{h}$ , 5-seed means).

| predictor        | skill | rank ic | note                                                               |
|------------------|-------|---------|--------------------------------------------------------------------|
| edge_persistence | 0.000 |         | spread today (reference)                                           |
| edge_ridge       | 0.192 | 0.67    | both endpoints’ day-ahead drivers + current spread (no graph)      |
| edge_gat         | 0.248 | 0.68    | GAT node embeddings → edge MLP head (message passing; 5-seed mean) |

The edge GAT beats the both-endpoint ridge by +0.056 skill — about 29% relative — in five seeds of five, and the synthetic control inverts as it should: on independent zones the same architecture is *worse* than the ridge, so the real-data gain is structure, not capacity. This is the project’s strongest relational result, and its placement is the point: node-level price forecasting barely needed the graph, congestion-as-edge-feature was null, and the GNN’s value concentrates precisely on the target that cannot be defined without the relation. Caveats stated once: a single six-month window, untuned hyperparameters, and modest absolute skill — spreads are noisy differences.

## 6.7 Summary of the energy evidence

Table 6.4 condenses the track. As an alpha project the energy track concluded negatively three artifacts deep, and the honest number — the strategy loses money — is stated as the finding it is. As a forecasting project it produced two robust, control-validated positives (the graph lift of

Table 6.4: Energy forecasting track: what is and is not supported (real ENTSO-E data, 20 zones, 5-seed means).

| finding                                      | comparison                | skill delta | seeds             | verdict  |
|----------------------------------------------|---------------------------|-------------|-------------------|----------|
| Graph helps price level                      | uniform - no-graph        | +0.131      | control-validated | robust   |
| Attention vs uniform (ranking)               | gat_rank_ic - uniform     | +0.031      | 5/5               | robust   |
| Attention vs uniform (skill)                 | gat - uniform             | -0.008      | 2/5               | wash     |
| Congestion edge feature (price-spread proxy) | gat_congestion - gat_node | -0.002      | 2/5               | null     |
| Congestion edge feature (real flow/NTC)      | flow - node               | -0.005      | 1/5               | null     |
| Edge-level spread (message passing)          | edge_gat - edge_ridge     | +0.056      | 5/5               | robust * |

+0.131 at node level; the edge-level message-passing gain of +0.056, five of five seeds) and one clean null (congestion edge features, under both operationalisations). As methodology it produced the report’s most transferable lessons: evaluate on untransformed returns, distrust magnitudes that outrun their plausibility class, and require cross-implementation replication before believing a learned effect.

## 7 Discussion

Every limitation below is a deliberate scope choice or a measured finding, not an oversight discovered in retrospect; several are results in their own right. This chapter names each threat to validity in the standard vocabulary and states where the project mitigated it, and where it simply lives with it.

**Look-ahead bias.** The report contains a full worked example of this threat rather than merely a disclaimer: the energy track’s first real-data result (Section 6.2) was look-ahead bias in two of its three layers, and the diagnostic arc that dismantled it is the chapter. On the equity track one mild instance is disclosed and retained: the static correlation graph is built from data up to the split date, so early *training* snapshots see a graph estimated partly from their own future. The out-of-sample window is unaffected — the graph strictly predates it — and the clean alternative, the dynamic graph, was rejected because it performs worse (Section 5.3). Disclosure was chosen over a cosmetic fix that costs real out-of-sample performance.

**Survivorship bias.** The equity universe is roughly fifty hand-picked, currently listed US large caps fetched from a free vendor. Names that failed or delisted over 2021–2026 are absent, which inflates the absolute level of every Sharpe in Chapter 5. The core claim is deliberately insulated: the attention A/B compares models on the *same* biased universe, so the bias raises both sides of the comparison rather than manufacturing the gap. Absolute levels should be read as optimistic; relative margins are the load-bearing numbers. A survivorship-bias-free vendor is the single highest-value data upgrade a successor project could make.

**Overfitting and data snooping.** Ten alphas, one composite, and repeated evaluation on one out-of-sample window invite snooping; the four gates exist to police exactly this, and the hyperparameter search was designed so the out-of-sample window could not influence selection (Section 5.5). The residual risks are stated rather than waved away: the walk-forward advantage is significant only within the tuned configuration ( $n = 5$  per arm) and is otherwise a directional claim; the flat

hyperparameter surface cuts both ways, as robustness and as a warning that tuning gains here are mostly noise; and the energy forecasting results rest on a single six-month window with untuned hyperparameters. The two synthetic negative controls and the shuffle-label control (Sections 5.1 and 6.1) are the strongest argument that the pipeline as a whole does not snoop.

**Transaction costs and tradability.** No result in this report is net of costs: there is no fill model, no spread, no market impact. The equity Sharpe of 1.37 is a paper Sharpe on daily bars, and a five-day-horizon long-short over fifty names would face real turnover costs. The energy track makes the point more brutally — its strategy loses money *before* any costs are applied (Section 6.4), and the E13b lesson (rank quality does not imply profitability when tails carry the money) would only sharpen under a cost model. The platform’s stated production boundary (Section 3.1) is consistent with this: nothing here is a trading system.

**Regime dependence and external validity.** All equity claims rest on a single 5.5-year window of one market regime, at one horizon ( $k = 5$  days), on one universe; all energy forecasting claims on six months of one calendar year, at one horizon ( $k = 24$  hours), on twenty zones. The attention analysis itself argues against over-generalising: the learned structure is temporally stationary (Section 5.6), so the project’s own evidence says the model found a stable statistical regularity, not a regime-adaptive mechanism. Whether either result survives a different half-decade is untested.

**Reproducibility.** What is reproducible is the distribution, not the point: the same seed produces different results on different devices, for identified mechanical reasons (Section 5.4). All headline numbers are therefore means over at least five seeds on a fixed device, the test suite pins the leakage-critical logic, and the archived per-run artifacts allow every table in this report to be regenerated. Bit-exact cross-device replication is not claimed and, on CUDA, not attainable.

**The open gate.** The relational composite never beat the best single island alpha — an OOS Sharpe of roughly 1.4 against roughly 3 — and this report resists the temptation to soften that bar post hoc. Two softer readings are available and honestly labelled as softer: the composite beats nine of its ten inputs, and it is nearly uncorrelated with all of them (maximum 0.26), which for a practitioner suggests portfolio value that the max-of-singles gate does not measure. A marginal-contribution-to-a-multifactor-portfolio reading is the natural next evaluation, planned but not executed within this project’s scope.

## 8 Conclusion and Future Work

The three research questions of Chapter 1 can now be answered directly.

*Does learned attention add value over unlearned relational propagation?* On the equity track, yes, with unusual consistency for a result in this domain: positive attention value-add in thirty of thirty seeded runs, across two hyperparameter configurations and two devices, against an anchor that shares the model’s inputs and topology. The mechanism is modest and now understood — a stationary, near-uniform reweighting of neighbour pooling — and the resulting composite is a real, unique, consistent factor that nevertheless does not beat the best single alpha it was built from. On the energy forecasting track the answer is narrower: attention improves cross-sectional ranking over the uniform anchor in five of five seeds but not mean-squared skill, where uniform pooling is already near-optimal.

*Does one kernel transfer across heterogeneous graphs without manufacturing false positives?* Yes, and the negative controls are the evidence: the same code that finds a consistent positive margin on real equity data finds nothing on synthetic random walks, finds nothing on synthetic uncoupled power zones, and — when real energy data produced numbers too good to be true —

the surrounding discipline treated the magnitude as an alarm rather than a result. The transfer claim is literal: one model class, one training loop, one gate suite, two structurally different graphs.

*Does the physical topology improve forecasts?* Yes, where it should. The interconnector graph lifts node-level price forecast skill by +0.131 over an identically-informed no-graph baseline, and the clearest relational gain sits on the irreducibly relational target: edge-level spread prediction beats a both-endpoint baseline by +0.056 skill in five of five seeds, while a synthetic control inverts the comparison. Congestion as an explicit edge feature added nothing robust under either of two operationalisations — an honest null.

Beyond the questions it set out to answer, the project’s most transferable output is methodological: evaluate on untransformed returns; treat rank correlation and profitability as different claims; treat implausible magnitudes as alarms; require seeds, fixed devices, and cross-implementation replication before believing a learned effect; and keep negative controls running permanently, because they are what give positive results their meaning.

## Future work

Four directions follow directly from the open ends documented in the experiment log and forecasting notes, in decreasing order of expected value. First, data: a survivorship-bias-free equity vendor would turn the biased absolute Sharpe levels into defensible ones, and longer, multi-window energy samples would test the six-month forecasting results out-of-sample in time. Second, evaluation: the value-added gate deserves its portfolio-level variant — the marginal contribution of the composite to a multifactor portfolio — alongside the max-of-singles bar it never cleared. Third, the forecasting track is untuned: a hyperparameter grid, seed ensembles, and walk-forward retraining, all already built for the alpha track, transfer mechanically; an edge-level congestion target (not just the spread) is the natural next prediction problem. Fourth, the congestion hypothesis remains open rather than dead: the flow-based-coupling regions publish shadow prices on critical network elements, and a congestion signal built from those — rather than from flow-over-capacity ratios with partial coverage — is the experiment this project would run next. Any renewed energy *alpha* attempt, by contrast, should wait for genuinely tradeable targets: balancing or intraday spreads, or transmission-rights-aware profit accounting.

The report closes where it began. The coupling structure of markets is information; on the evidence here, learning how to read it earns a consistent but modest premium over reading it naively, and the discipline required to measure that premium honestly — anchors, controls, embargoes, seeds, and the willingness to let three spectacular numbers die — is itself the result this project most confidently recommends to its successors.



# Bibliography

- [1] Z. Kakushadze, *101 formulaic alphas*, 2016. arXiv: 1601.00991 [q-fin.PM]. [Online]. Available: <https://arxiv.org/abs/1601.00991>
- [2] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, *Graph attention networks*, 2018. arXiv: 1710.10903 [stat.ML]. [Online]. Available: <https://arxiv.org/abs/1710.10903>
- [3] S. Gu, B. Kelly, and D. Xiu, “Empirical asset pricing via machine learning,” *The Review of Financial Studies*, vol. 33, no. 5, pp. 2223–2273, 2020. DOI: 10.1093/rfs/hhaa009
- [4] M. López de Prado, *Advances in Financial Machine Learning*. Hoboken, NJ: Wiley, 2018, ISBN: 978-1-119-48208-6.
- [5] T. N. Kipf and M. Welling, *Semi-supervised classification with graph convolutional networks*, 2017. arXiv: 1609.02907 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1609.02907>
- [6] S. Brody, U. Alon, and E. Yahav, *How attentive are graph attention networks?* 2022. arXiv: 2105.14491 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/2105.14491>
- [7] M. Fey and J. E. Lenssen, *Fast graph representation learning with PyTorch Geometric*, 2019. arXiv: 1903.02428 [cs.LG]. [Online]. Available: <https://arxiv.org/abs/1903.02428>
- [8] F. Feng, X. He, X. Wang, C. Luo, Y. Liu, and T.-S. Chua, *Temporal relational ranking for stock prediction*, 2019. arXiv: 1809.09441 [cs.CE]. [Online]. Available: <https://arxiv.org/abs/1809.09441>
- [9] R. Weron, “Electricity price forecasting: A review of the state-of-the-art with a look into the future,” *International Journal of Forecasting*, vol. 30, no. 4, pp. 1030–1081, 2014. DOI: 10.1016/j.ijforecast.2014.08.008
- [10] J. Lago, G. Marcjasz, B. De Schutter, and R. Weron, “Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark,” *Applied Energy*, vol. 293, p. 116983, 2021. DOI: 10.1016/j.apenergy.2021.116983
- [11] ENTSO-E. “ENTSO-E Transparency Platform.” Day-ahead prices, load and renewable forecasts, physical cross-border flows; accessed via the RESTful API, Accessed: 2026-07-02. [Online]. Available: <https://transparency.entsoe.eu/>
- [12] M. Raasveldt and H. Mühleisen, “DuckDB: An embeddable analytical database,” in *Proceedings of the 2019 International Conference on Management of Data (SIGMOD '19)*, 2019. DOI: 10.1145/3299869.3320212
- [13] dbt Labs. “dbt: Transform data in your warehouse,” Accessed: 2026-07-02. [Online]. Available: <https://www.getdbt.com/>

# A Reproducibility

Everything claimed in this report can be regenerated from the repository.<sup>1</sup> This appendix records the environment, the commands, and the mapping from experiment to archived artifact.

## Environment

Python 3.13 with `torch 2.12.0` and `torch_geometric` (installed via the project’s `[gnn]` extra); `PYTHONPATH` set to `src`. All headline experiments ran on CPU — a deliberate choice after the device study of Section 5.4 — with the device recorded per experiment. The equity track fetches market data through `yfinance`; the energy track runs on the synthetic generator by default and on live ENTSO-E data when an API token is supplied via the `ENTSOE_API_KEY` environment variable. The warehouse and marts require `duckdb`, `dbt-core`, and `dbt-duckdb`; the GAT pipeline itself does not.

## Tests

The full suite is 115 tests; 55 of them cover the capstone modules (graph construction, training primitives, propagators, factor providers, both track runners, attention analysis, and the leakage controls). The leakage controls — shuffle-label negative control and planted-signal positive control — are pinned to CPU for determinism and run on every change. `py -m pytest` from the repository root executes everything.

## Experiments and artifacts

Each experiment entry in the log (`docs/gat_experiment_log.md`, entries E1–E14) names its run script and its archived outputs. The run scripts are one-off drivers kept in an untracked scratch area outside the published repository; each log entry records the configuration they ran — model variant, seeds, hyperparameters, device — and the CLI commands below drive the same pipelines. The outputs — per-run diagnostics, matrix, seed, hyperparameter-grid, winner-validation, energy, and attention CSV/JSON files, plus five attention figures — are date-prefixed under `docs/results/`. The canonical trained equity weights are tracked at `data/warehouse/gat_equity.pt`. Every quantitative table in this report is generated mechanically from those archived CSVs; no result number in the text was typed by hand from memory.

## Command reference

The equity pipeline:

```
quant-alpha gat-equity --graph {static,dynamic}
                        --retrain {single,walk-forward}
                        --loss {ic,mse}
                        --device {cpu,cuda,auto}
                        --persist
```

---

<sup>1</sup><https://github.com/witold-andelie/quant-foundation-gat>

runs end to end and, with `-persist`, writes the four warehouse tables of Section 3.3. The energy alpha runner is `run_gat_energy` (synthetic or `source="entsoe"`); the forecasting ladder is `quant-alpha energy-forecast -source {synthetic,entsoe} -k 24` with a `-raw-path` Parquet cache for repeatable real-data runs. The marts rebuild with `dbt build -profiles-dir .` inside `dbt_quant_alpha/` (21 models and tests) and `dbt_energy_alpha/` (12). The report's build script regenerates the tables and figures from the repository's artifacts and reference data and then runs `pdflatex` and `biber`.

## What reproducibility means here

As established in Section 5.4, bit-exact replication across devices is not attainable for CUDA-adjacent training stacks and is not claimed. Rerunning any experiment with the recorded seeds on a CPU of the same class reproduces the reported distributions — means and standard deviations over five or more seeds — and every individual archived run is identified by seed, device, configuration, and date in its artifact file.